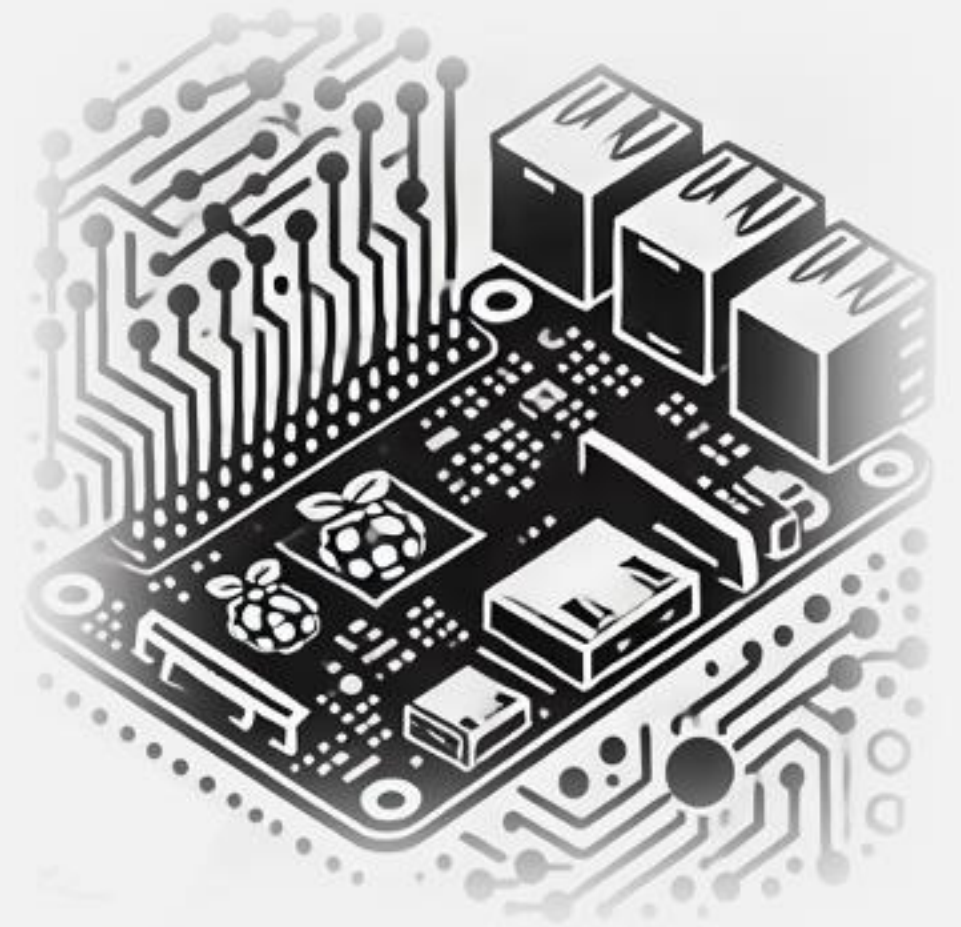


IESTI05 – Edge AI

Machine Learning System Engineering

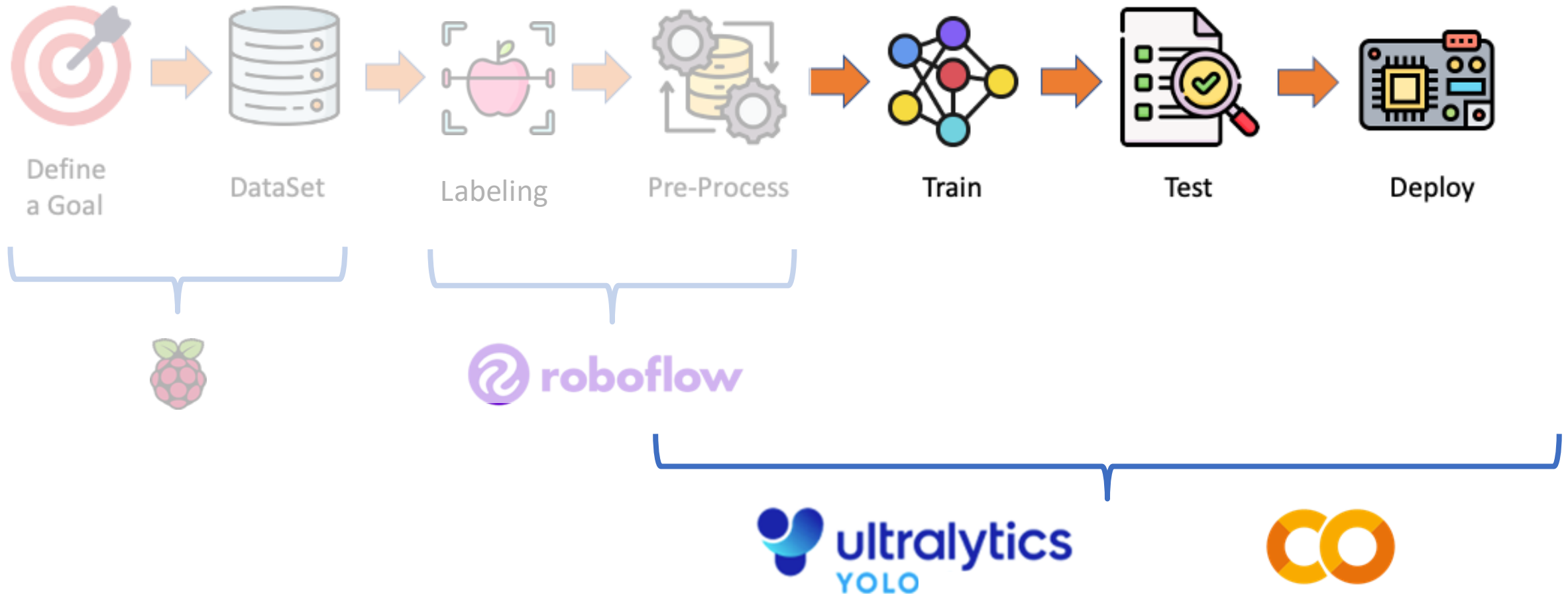
12.Object Detection Project: Train & Deploy – YOLO



The Goal



Machine Learning Workflow



Data Aquisition



Universe

Search 500,000+ Open Source Computer Vision Projects...

Back

Box versus Whe...
Object Detection

Versions

v8 2025-09-01 2:56pm
Generated on Sep 1, 2025

Download Dataset

Use this Dataset

Use this Model

Download

Image and Annotation Format

YOLOv11

TXT annotations and YAML config used with YOLOv11.

Download Options

☐ Download zip to computer
Downloads all images, annotations, and classes.

☒ Show download code
Custom train this dataset using the provided code snippet in a notebook.

Cancel

Continue

248 Total Images

View All Images →

Dataset Split

TRAIN SET 91%

VALID SET 5%

TEST SET 4%

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Object Detection

GENERAL

Overview

DATA

Images 98

Dataset

Analytics

DEPLOY

Model 2

API Docs

Versions

Use this Dataset Use this Model

Download Dataset

v8 2025-09-01 2:56pm
Generated on Sep 1, 2025

2025-09-01 2:56pm
v8 248 320x320
Stretch

2025-09-01 2:56pm
v7 248 320x320
Stretch

Roboflow

2024-09-01 2:56pm
v6 248 320x320
Stretch

2024-09-01 2:56pm
v5 248 320x320
Stretch

2024-09-01 2:56pm
v4 248 320x320
Stretch

2024-04-14 3:01pm
v3 126
Stretch

2024-04-12 3:19pm
v2 126 96x96
Stretch

Dataset Split

TRAIN SET	91%	VALID SET	5%	TEST SET	4%
225 Images		13 Images		10 Images	

Preprocessing

Auto-Orient: Applied
Resize: Stretch to 320x320

Augmentations

Outputs per training example: 3

Download

Jupyter Terminal Raw URL

Paste this snippet into a notebook from our model library to download and unzip your dataset:

```
!pip install roboflow

from roboflow import Roboflow
rf = Roboflow(api_key="")
project = rf.workspace("marcelo-rovai-riilo").project("box-versus-wheel-auto-dataset")
version = project.version(8)
dataset = version.download("yolov11")
```

Done Choose a Model

COPY

View All Images →

Ultralytics Training:
[yolo11_box_vs_wheel.ipynb](#)

[\[OPEN COLAB\]](#)



yolo11_box_vs_wheel.ipynb

https://colab.research.google.com/drive/1tvVtXFoKIZBT1RFbpq8C9ODjygBBuVuS#scrollTo=bSq6KopwNxnF

CO

yolo11_box_vs_wheel.ipynb

File Edit View Insert Runtime Tools Help

Commands Code Text Run all

RAM Disk

Files

datasets

Box-versus-Wheel-auto-datase..

test

images

labels

train

images

labels

labels.cache

valid

README.dataset.txt

README.roboflow.txt

data.yaml

Disk 73.16 GB available

Dataset

[5]

1 !mkdir {HOME}/datasets

2 %cd {HOME}/datasets

3

4 !pip install roboflow --quiet

5

6 from roboflow import Roboflow

7 rf = Roboflow(api_key="8NEOn2tX76EJbk9hQTb4")

8 project = rf.workspace("marcelo-rovai-riila").project("box-versus-wheel-auto-dataset")

9 version = project.version(8)

10 dataset = version.download("yolov11")

11

/content/datasets

88.6/88.6 kB 5.2 MB/s eta 0:00:00

66.8/66.8 kB 6.8 MB/s eta 0:00:00

49.9/49.9 MB 20.3 MB/s eta 0:00:00

1.4/1.4 MB 75.1 MB/s eta 0:00:00

4.2/4.2 MB 108.1 MB/s eta 0:00:00

loading Roboflow workspace...

loading Roboflow project...

Downloading Dataset Version Zip in Box-versus-Wheel-auto-dataset-8 to yolov11:: 100%|██████████| 4855/4855 [00:

Extracting Dataset Version Zip to Box-versus-Wheel-auto-dataset-8 in yolov11:: 100%|██████████| 508/508 [00:00:

Variables

Terminal

2:55 PM

T4 (Python 3)

PAST

7

yolo11_box_vs_wheel.ipynb

https://colab.research.google.com/drive/1tvVtXFoKIZBT1RFbpq8C9ODjygBBuVuS#scrollTo=cK5WknXSVOoN

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

Files

- datasets
- gdrive
- runs
 - detect
 - predict
 - train
 - weights
 - best.pt
 - last.pt
 - BoxF1_curve.png
 - BoxPR_curve.png
 - BoxP_curve.png
 - BoxR_curve.png

Export the Training Result to the Gdrive

```
[15] 1 from google.colab import drive
      2 drive.mount('/content/gdrive')

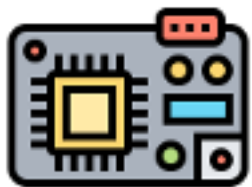
Mounted at /content/gdrive

[ ] 1 /content/gdrive/MyDrive/10_UNIFEI/IESTI05-Box-Wheel-2025

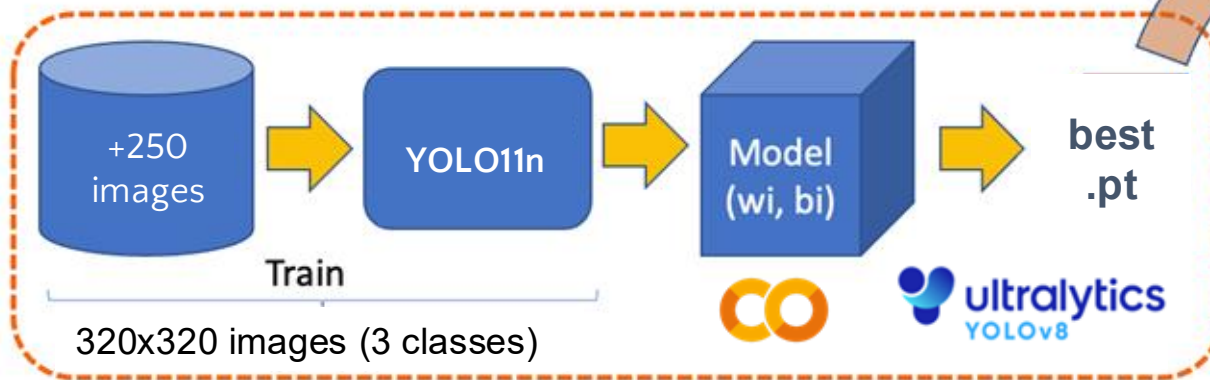
[16] 1 !scp -r /content/runs '/content/gdrive/MyDrive/10_UNIFEI/IESTI05-Box-Wheel-2025'
```

box 0.94 wheel 0.95

2:55 PM T4 (Python 3)



Ultralytics YOLO11n



5.5 MB



Raspberry Pi Inference:

[YOLO Model Predictions with Ultralytics.ipynb](#)



Questions?



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rovai@unifei.edu.br



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